S11 OPTIMIZER-INFORMATION ABLATION

In the following experiment, we are testing ablation on the validation data only. To do so, we run the discovery loop with 5 levels of information: starting with the full workspace, to having only the harness as a signal (Table S11.1). We use `gpt-5.6-luna` for this, due to cost. We use medium reasoning effort for both the detection and optimizer agents. We run 5 iterations for each of these 5 levels. This means the agent has 5 iterations to improve the harness, having only access to that "level" of information as the signal. All levels start from the same seed harness, and only differ by what the level's signals contain. The *Optimizer workspace* column reveals what the workspace has access to. For `no-labels`, fields whose position reveals the label were also removed, and sample identifiers were replaced by opaque hashes.

Overall, our findings are as follows. All levels improved significantly over the seed (from 9.9 to between 19.8 and 23.3 pairwise accuracy, exact McNemar $p \leq 0.005$), and no level differed significantly from another (all $p \geq 0.33$). We consider this ablation inconclusive: it uses a single run per level, a single optimizer configuration (`gpt-5.6-luna`, an efficiency-oriented model, at medium reasoning effort) that differs from those in our main experiments, and validation data only.

Level	Optimizer workspace	It. 0	It. 1	It. 2	It. 3	It. 4	It. 5	Best	Mean
`full`	harnesses, metrics, history, predictions with labels	9.90	19.80	14.36	18.81	13.37	17.33	19.80	16.73
`no-labels`	same, labels and label-revealing fields removed	9.90	16.83	19.80	19.80	20.30	17.82	20.30	18.91
`no-predictions`	harnesses, metrics, history	9.90	21.78	16.83	19.31	19.31	19.31	21.78	19.31
`no-history`	latest metrics only	9.90	16.83	19.31	21.29	23.27	19.80	23.27	20.10
`harness-only`	current harness only	9.90	16.34	20.79	19.31	15.84	21.29	21.29	18.71

Table S11.1: Optimizer-information ablation on the validation set (202 pairs, 404 functions), **pairwise accuracy** (%). All levels start from the same seed harness (It. 0). Mean is over iterations 1–5.